

04-July-2026

Agenda:

ML-14

→ Support Vector Machine (SVM)

→ primal form (linear SVM) → completed

→ Dual form SVM •

→ Regression → SVR •

→ practicals (SVM + gridsearchcv)

$$J(w, b) = \frac{1}{2} \|w\|^2 + c \sum_{i=1}^n \max(0, 1 - y_i (w \cdot x_i + b))$$

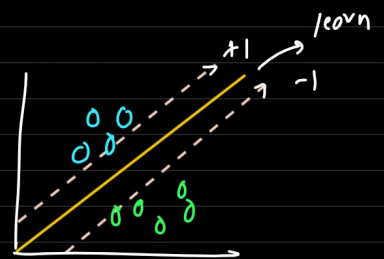
path A: ($z \geq 1$), loss $\rightarrow 0$

$$\frac{\partial J}{\partial w} = w \quad \frac{\partial J}{\partial b} = 0$$

path B: ($z < 1$), loss is active

$$\frac{\partial J}{\partial w} = w - (c \cdot y_i \cdot x_i)$$

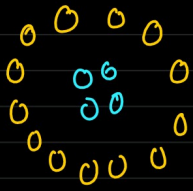
$$\frac{\partial J}{\partial b} = -(c \cdot y_i)$$



The problem: when straight line fails

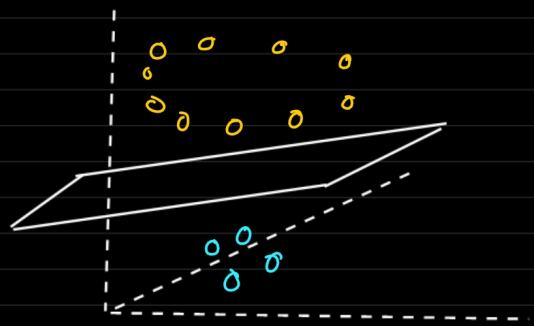
linear regression $\rightarrow$ polynomial regression





2D

transform the data
into higher dimension



(Dual form - kernelized SVM)

$$A: [2, 3] \rightarrow$$

2D

(polynomial function)

$$\phi(x) =$$

$d=2$

$$[x_1^2, \sqrt{2}x_1x_2, x_2^2]$$

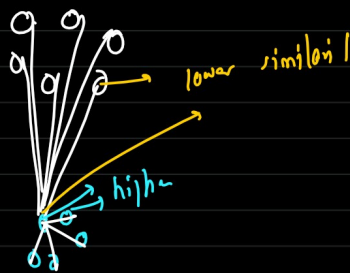
$$A: [2^2, \sqrt{2(2)(3)}, 3^2]$$

$$[4, 6\sqrt{2}, 9]$$

3D

$$B: [4, 1]$$

$$[16, 4\sqrt{2}, 1]$$



Requirement:

1) mathematical function $\rightarrow$ high dimensional data

2) close or similar to their class $\rightarrow$ dot product

$$A: [4, 6\sqrt{2}, 9] \quad B: [16, 4\sqrt{2}, 1]$$

→ Take the 3D dot product:

$$(4 \times 16) + (6\sqrt{2} \times 4\sqrt{2}) + (9 \times 1) = 64 + (24 \times 2) + 9 \\ = 121$$

2D → 3D → dot product → single value ← similarity score

A-B

closer → higher score

far → lower score

2D → 3D

100,000

3D

□ □ □

→ memory
→ computation

100,000 × 3

computation

dot product score

$$\rightarrow (A \cdot B)^2$$

step-1 → A → 3D

step-2 → B → 3D

step-3 → A dot B

$$\left(\begin{bmatrix} 2, 3 \end{bmatrix} \cdot \begin{bmatrix} 1, 1 \end{bmatrix} \right)^2 \rightarrow \left((2 \times 1) + (3 \times 1) \right)^2 \rightarrow \left((2) + (3) \right)^2 \\ \rightarrow (5)^2 \rightarrow 25$$

kernel function

or

kernel trick

$$A - B \rightarrow \square$$

$$A - C \rightarrow \square$$

computer

3D

$$\text{car} \rightarrow (\text{red}, 240 \text{ km/hr}, 40) \rightarrow A$$

$$\text{car} \rightarrow (\text{yellow}, 120 \text{ km/hr}, 20) \rightarrow B$$

$$\text{car} \rightarrow (\text{yellow}, 230 \text{ km/hr}, 35) \rightarrow C$$

$$\text{car} \rightarrow (\text{blue}, 110 \text{ km/hr}, 5) \rightarrow D$$

$$\text{car} \rightarrow (\text{green}, 130 \text{ km/hr}, 10) \rightarrow \square$$

3D

B, C

2D $\rightarrow$ ND

linear kernel:

SVM $\rightarrow$ kernel $\rightarrow$ linear

$$\phi(x) = x$$

$$\phi(2,3) = [2, 3]$$

$$\phi(2,3)$$

$$\phi(4,1)$$

A · B

polynomial kernel:

$$\phi(x) : [x_1^2, \sqrt{2}x_1x_2, x_2^2]$$

$d=2$

$$\phi(2,3) : [4, 6\sqrt{2}, 9] \rightarrow (A \cdot B)^2$$

$$\phi(4,1) : [16, 4\sqrt{2}, 1]$$

RBF kernel (Radial Basis function / Gaussian)

maps data to infinite dimension

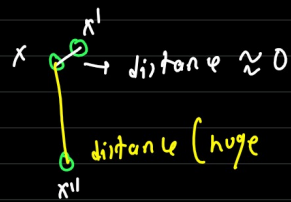
$\rightarrow$ Taylor series $\rightarrow e^x \rightarrow$ polynomial goes forever

$$k(x, x') = \exp(-\gamma \|x - x'\|^2)$$

$\rightarrow$ square euclidean distance between two point

data $\rightarrow$ infinite $\rightarrow$ kernel $\rightarrow$

instead of thinking about infinite dimension, think about close & far points are



point right on top of each other
→ distance 0

→ distance 0

$$\rightarrow \exp(0) \rightarrow 1$$

point that are miles apart

→ distance huge

$$\rightarrow \exp(-h\nu_e) \approx 0$$

RBF $\rightarrow$ similarity score ranging from 1 (identical) to 0 (unrelated)



→ 10 → 20 → 5 → 9
Hello, How are you → ① [10, 20, 5, 9]

Hallo, are you How $\rightarrow$ (2) $[10, 5, 4, 20]$

Hello $\rightarrow$ (0 0 0 0 0 0 0) $\xrightarrow{1 \times 512}$
 Hello. $\rightarrow$ (0 0 0 0 0 0 0) $\xrightarrow{1 \times 512}$

SVM:

we want to minimize, subject to

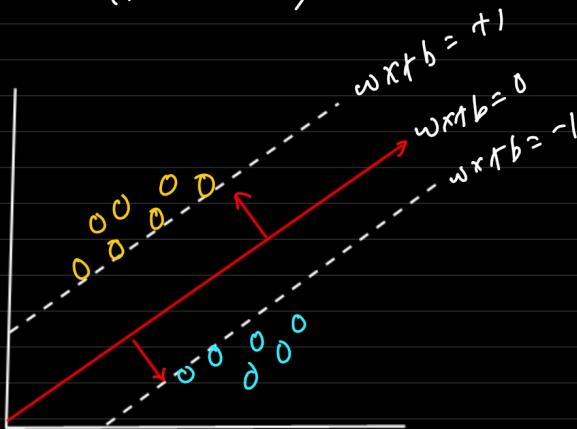
$$\frac{1}{2} \|w\|^2$$

$$y_i (w \cdot x_i + b) \geq 1$$

$$Z \geq 1 \rightarrow \text{no loss}$$

$$Z < 1 \rightarrow \text{learning}$$

SVM (primal form)



learnable parameters: w, b

formula +
dot-product $\rightarrow$ single score

$\downarrow$
kernel

SVM (dual form)



idea:

$\rightarrow$ increase the dimension of the data to make them separable.

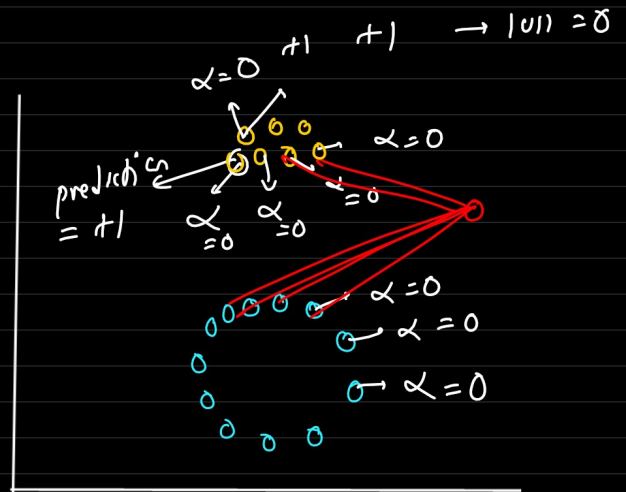
$\downarrow$
2D $\rightarrow$ 3D
 $x_1, x_2 \quad x_1, x_2, x_3$

$\downarrow$ 2D $\rightarrow$ 4D
 $\downarrow$ 2D $\rightarrow$ 5D } issue: computation

$\downarrow$

student $\rightarrow$ Age, class, height, score(A, B, C, D, E)
(#)(8) $\rightarrow$

(B)(8) $\rightarrow$



→ Dual form, the kernel is strictly a pair-wise operation

prediction $\leftarrow$ formula [voting from all pairs]
 $= +1$

Actual $\rightarrow$ (OS) $\rightarrow$

10,000 data $\rightarrow$ learnable param $\rightarrow \alpha_{total} \rightarrow 10,000$

$\rightarrow \alpha_{total} \rightarrow 50 - 100$

→ kernel ✓

→ primal & dual form

↓
 → kernel

→ similarity concept

$2(w)$

$x_1 \ x_2$

2D $\rightarrow$

$3(w)$

$x_1 \ x_2 \ x_3$

3D

RBF

$3(\infty)$

2D $\rightarrow$ ND

prediction $\rightarrow y_i(\omega \cdot x + b)$ (primal form)

prediction $\rightarrow \sum_{i=1}^n \alpha_i y_i (k(x_i, x) + b)$ (dual form)

Iteration 1: $d=2, b=0, \alpha=0.01$

$$k(x, x') = (x \cdot x')^2$$

precomputed similarities

→ select a data point

→ A

$x_1 \quad x_2 \quad y_A$
A 2 3 +1 → training loop 1:

B 4 1 -1

C 0 1 +1

$$\alpha_A \alpha_B \alpha_C = 0$$

$$k(A, A) = ((2 \times 2) + (3 \times 3))^2 = 13^2 = 169$$

$$k(B, A) = ((4 \times 2) + (1 \times 3))^2 = 11^2 = 121$$

$$k(C, A) = ((0 \times 2) + (1 \times 3))^2 = 3^2 = 9$$

→ step 2 (prediction A) $\sum_{i=1}^n \alpha_i y_i (k(x_i, x) + b)$

vote from A: $\alpha_A \cdot y_A \cdot k(A, A) = 0.1 \cdot 169 = 0$

vote from B: $\alpha_B \cdot y_B \cdot k(B, A) = 0.1 \cdot 121 = 0$

vote from C: $\alpha_C \cdot y_C \cdot k(C, A) = 0.1 \cdot 9 = 0$

Total-prediction for point A = vote(A + B + C)
= 0.0

step 3 → Check Error (Hinge loss)

$z = y_A \cdot \text{prediction}$

$z = (1) \cdot (0) = 0.0$

$z < 1 \leftarrow$ Error, update learning rate

$\max(0, 1 - z)$ primal form
 $\max(0, 1 - y_i (w \cdot x_i + b))$
 $\max(0, 1 - y_i (\alpha_i y_i (k(x_i, x) + b)))$
 $\max(0, 1 - z)$ dual form

step 4: update learning rate

$\alpha_A = \alpha_A + \text{learning rate}$

$= 0 + 0.01 = 0.01$

Step-5 : Re-predict point A

$$\text{vote-A} : \alpha_A \cdot y_A \cdot k(A, A) = 0.01 \cdot 1 \cdot 1.69 = 1.69$$

$$\text{vote-B} : 0$$

$$\text{vote-C} : 0 \\ = 1.69$$

$$z = y_i \cdot \text{prediction} \rightarrow (1)(1.69) \rightarrow 1.69$$

$z > 1 \rightarrow$ point is in safe zone

Iteration: 2

Step-1 compute similarity

	x_1	x_2	y_A	α
A	2	3	+1	0.01
B	4	1	-1	0
C	0	1	+1	0

$\rightarrow$ training

$$k(A, B) = 121$$

$$k(B, B) = ((4 \times 4) + (1 \times 1))^2 = 289$$

$$k(C, B) = ((0 \times 4) + (1 \times 1))^2 = 1$$

Step-2 : prediction B

$$\text{vote from A} : \alpha_A \cdot y_A \cdot k(A, B) = 0.01 \cdot (1) \cdot 121 = 1.21$$

$$\text{vote from B} : \alpha_B \cdot y_B \cdot k(B, B) = 0 \cdot (-1) \cdot 289 = 0$$

$$\text{vote from C} : \alpha_C \cdot y_C \cdot k(C, B) = 0 \cdot (1) \cdot 1 = 0$$

$$\text{Total predicted} = \text{vote (A+B+C)} \\ = 1.21 \rightarrow$$

vote from A pulled is
telling that A & B are
similar

Step:3 $\rightarrow$ Error

$$z = y_B \cdot \text{prediction} \rightarrow (-1)(1.21) = -1.21$$

$-1.21 > 1 \rightarrow$ Massive Error

step: 4 learning

$$\alpha_\beta = \alpha_\beta + \text{learning} \\ = 0.0 + 0.01 \rightarrow 0.01$$

step: 5 evaluate

$$\text{vote-A} : \alpha_A \cdot y_A \cdot k(A, B) = 0.01 \cdot 1 \cdot |21| = 1.21$$

$$\text{vote-B} : 0.01 \cdot (-1) \cdot 289 = -2.89$$

$$\text{vote-C} : 0$$

$$= 1.21 - 2.89 + 0 \rightarrow -1.68$$

$$z = y_i \cdot \text{prediction} \rightarrow (-1)(-1.68) \rightarrow 1.68$$

$z > 1 \rightarrow$ point is in safe zone

$$\text{previous } z \rightarrow -2.89$$

$$\text{new } z \rightarrow 1.68 > 1 \rightarrow \text{safe}$$

primal form

learnable
params

weight (w) and bias (b)

prediction

$$\text{prediction} = w \cdot x + b$$

confidence

$$z = y \cdot \text{prediction}$$

learning
flag

Is $z < 1$?
 $\rightarrow$ update

Dual form

temporale score (α) for each point
& bias (b)

$$\text{prediction} = \sum \alpha_i y_i k(x_i, x) + b$$

$$z = y \cdot \text{prediction}$$

Is $z < 1$?
 $\rightarrow$ update

update $w_{new} = w_{old} - \alpha (\text{gradient } f)$
Math

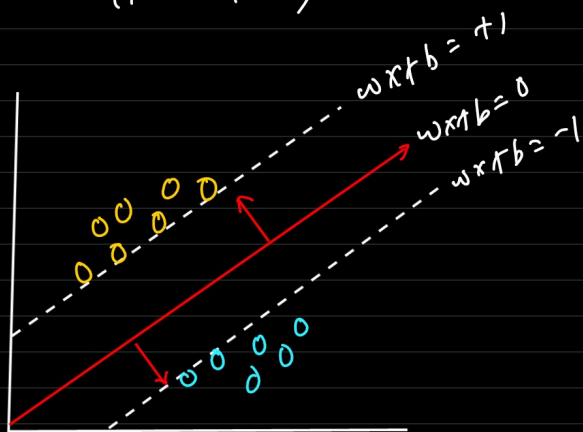
$\alpha_{new} = \alpha_{old} \times \text{learning rate}$

mental
model tilt & shift a rigid 2D
highway (centre line & margin)

Discovering boundary point, via weighted
similarity search

Support vector regression

SVM (primal form)



learnable parameters: w, b

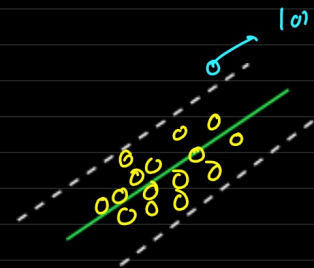
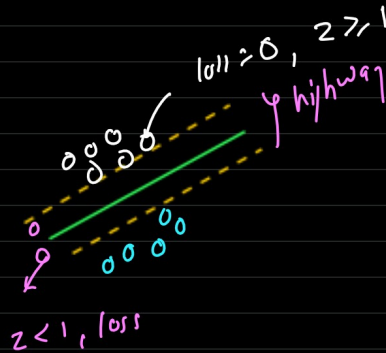
boundary

something that not only follows the general loss, also something that is
least affected by outlier.

remove the square from the loss

$$J(w, b) = \frac{1}{2} \|w\|^2 + c \sum_{i=1}^n \max(0, 1 - y_i (w \cdot x_i + b))$$

$$+ c \sum_{i=1}^n \max(0, |y_i - y_{\text{pred}}| - \epsilon)$$

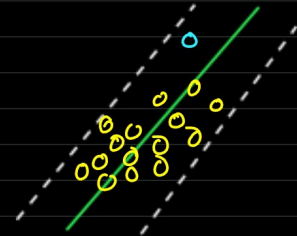


Inside rule: $|y_i - y_{\text{pred}}| \leq \epsilon$

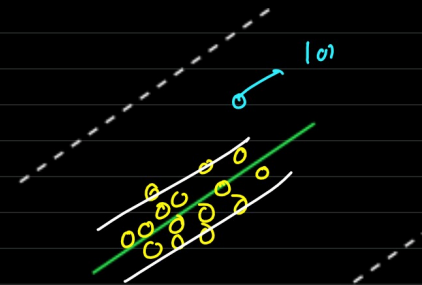
unit of target column

height

$y_i - y_{\text{pred}} \leq 10 \text{ cm}$



$\epsilon \rightarrow$ margin of acceptable error



	Height	$\hat{y}$	$ y - \hat{y} $	$\epsilon = 2$	$\epsilon = 5000$
A	171	172	1	→	→
B	181	200	19	→	→
C	191	180	11	→	→
D	200	60	140	→	→
E	169	170	6	→	→



SVM classification

→ point is inside the margin $z < 1$

→ point is outside the margin $z > 1$

SVM regression:

→ point too high (C)

→ point too low (B)

→ point is inside (A)

path A: $|y_{true} - y_{pred}| \leq \epsilon$

$$\frac{\partial J}{\partial w} = w \quad (loss = 0)$$

$$\frac{\partial J}{\partial b} = 0$$

path B: prediction is too low $(y_{true} > y_{pred} + \epsilon)$

$$180 (> 160 + 10)$$

$$\frac{\partial J}{\partial w} = w - (C \cdot x_i)$$

$$180$$

$$\frac{\partial J}{\partial b} = -C$$

path C: prediction too high $(y_{true} < y_{pred} - \epsilon)$

$$180 < (200 - 10)$$

$$\frac{\partial J}{\partial w} = w + (C \cdot x_i)$$

$$\frac{\partial J}{\partial b} = +C$$

Iteration - 1

x (k of sq ft)	y (hundred of thousand)	$\epsilon = 2$	$\alpha = 0.1$
5	12	$C = 1.0$	$w = 1.0, b = 1.0$

step 1: predict the price

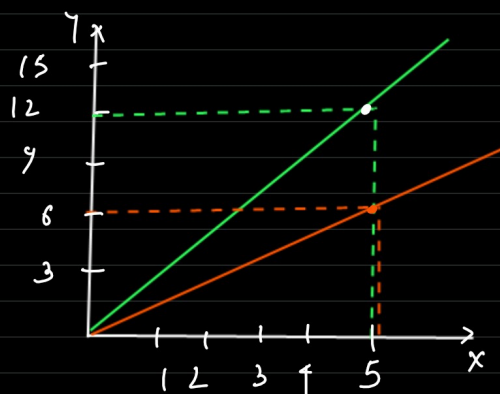
$$\begin{aligned} y_{\text{pred}} &= (w \cdot x) + b \\ &= (1.5) + 1 = 6.0 \end{aligned}$$

step 2: check the error $|y_{\text{true}} - y_{\text{pred}}| = (12.0 - 6.0) = 6.0$

$$\begin{aligned} \text{error} &\leq \epsilon \\ \rightarrow 6.0 &\leq 2.0 \rightarrow \end{aligned}$$

actual $\rightarrow 12.0$

The prediction is too low $\rightarrow$ path B



step 3: calculate gradient (path B)

$$\text{Gradient}_w = w - (C \cdot x) = 1.0 - (1.5) = -4.0$$

$$\text{Gradient}_b = -C = -1.0$$

step 4: update weight

$$w_{\text{new}} = w_{\text{old}} - \alpha (\text{gradient}) = 1.0 - 0.1 (-4.0) = 1.0 + 0.4 = 1.4$$

$$b_{\text{new}} = b_{\text{old}} - \alpha (\text{gradient}) = 1.0 - 0.1 (-1.0) = 1.0 + 0.1 = 1.1$$

steps: Evaluate / pred

$$y_{\text{pred-new}} = (w \cdot x + b) = (1.4 \cdot 5) + 1.1 \rightarrow 7.0 + 1.1 \rightarrow 8.1$$

$$\begin{array}{lll} y_{\text{pred-old}} = 6.0 & |y_{\text{true}} - y_{\text{pred}}| = 6.0 & 6.0 \leq 2.0 \\ y_{\text{pred-new}} = 8.1 & |y_{\text{true}} - y_{\text{pred}}| = 8.1 - 12 & 3.9 \leq 2.0 \end{array} \quad \downarrow \text{improve}$$

SVR (Dual-form)

→ kernel

$$\text{prediction} = \sum_{i=1}^n (\alpha_i^* - \alpha_i) \cdot k(x_i, x) + b$$

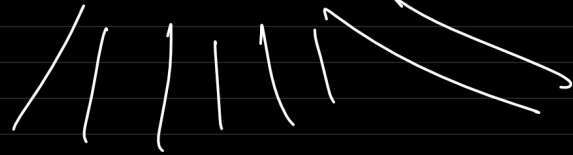
α_i^* → Alpha-star
 power to pull the prediction up
 α_i → Regular-Alpha
 power to pull the prediction down



$0.1, x, \gamma$



$0.1, x', \gamma'$



100

→ 40

